

# Cancer Clinical Trials & Evidence-Based Oncology

Trial Design, Endpoints, Statistical Concepts, and Interpreting Oncology Data

Doc #15 of 15

Domain: Oncology & Cancer Treatment

Audience: Medical Professionals

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## 1. Clinical Trial Phases

| Phase      | Primary Objective                     | Typical N | Key Questions                                                          |
|------------|---------------------------------------|-----------|------------------------------------------------------------------------|
| Phase 0    | Pharmacokinetics/pharmacodynamics     | <15       | Does drug reach target? Correct dose range?                            |
| Phase I    | Maximum tolerated dose (MTD) / safety | 15-50     | What dose is safe? What is the DLT? PK parameters?                     |
| Phase I/II | Dose expansion / preliminary efficacy | 50-150    | RP2D; early signals of efficacy in biomarker-selected cohorts          |
| Phase II   | Preliminary efficacy; signal-finding  | 50-200    | ORR, PFS, safety; single arm or randomised; go/no-go decision          |
| Phase III  | Definitive efficacy vs. control       | 200-3000+ | Superiority or non-inferiority vs. standard of care; OS or PFS primary |
| Phase IV   | Post-marketing surveillance           | 1000+     | Long-term safety; rare adverse events; real-world effectiveness        |

## 2. Key Endpoints in Oncology Trials

| Endpoint                        | Definition                                                     | Regulatory Status                                 | Caveat                                                                         |
|---------------------------------|----------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------|
| Overall Survival (OS)           | Time from randomisation to death from any cause                | Gold standard; FDA full approval                  | Requires large N and long follow-up; confounded by post-progression therapy    |
| Progression-Free Survival (PFS) | Time from randomisation to progression or death                | Accepted surrogate; used for accelerated approval | Correlation with OS varies by tumour type (strong in MM, ovarian; weak in CRC) |
| Overall Response Rate (ORR)     | Proportion achieving CR + PR (RECIST 1.1)                      | Single-arm basis for accelerated approval         | No duration information; must be durable to be clinically meaningful           |
| Duration of Response (DOR)      | Time from first response to progression                        | Contextualises ORR                                | Applicable only to responders; subject to selection bias                       |
| Disease Control Rate (DCR)      | CR + PR + SD                                                   | Exploratory                                       | SD definition (>6w vs >12w) varies across trials                               |
| Event-Free Survival (EFS)       | Time to relapse, progression, or death — used in adjuvant/haem | Accepted in haematology                           | Composite; definition varies                                                   |
| pCR (pathological CR)           | No residual invasive cancer post-neoadjuvant treatment         | Surrogate endpoint (breast, rectal)               | Correlation with OS requires validation per tumour type                        |

### 3. Essential Statistical Concepts

- Hazard Ratio (HR): Ratio of event rates between treatment and control arms in time-to-event analysis. HR <1 favours treatment (e.g., HR 0.70 = 30% reduction in hazard of the event). Constant HR assumes proportional hazards — violated in some immunotherapy trials (delayed separation of curves)
- Confidence Interval (CI): 95% CI that does not cross 1.0 (for HR) or 0 (for difference) indicates statistical significance at  $p < 0.05$
- Log-rank test: Standard method to compare survival curves; most powerful when hazards are proportional
- RMST (Restricted Mean Survival Time): Alternative to HR when proportional hazards assumption is violated (relevant in ICI trials with delayed effects and 'tail' benefit)
- Power and sample size: Typically 80-90% power; Type I error ( $\alpha$ ) 5% (two-sided). Under-powered trials risk false-negative results
- Intention-to-treat (ITT) analysis: All randomised patients analysed in assigned groups regardless of actual treatment received — preserves randomisation; the primary analysis for Phase III trials
- Per-protocol (PP) analysis: Only patients who completed therapy as specified — may inflate efficacy estimate; used as sensitivity analysis

### 4. Modern Adaptive & Biomarker-Enriched Trial Designs

| Design | Description | Oncology Example |
|--------|-------------|------------------|
|--------|-------------|------------------|

|                       |                                                                                                |                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Basket Trial          | Single agent tested across multiple tumour types sharing a molecular alteration                | KEYNOTE-158 (pembro in TMB-H); larotrectinib in NTRK+ (LOXO-TRK-14001) |
| Umbrella Trial        | Multiple targeted therapies tested within one tumour type based on molecular subtypes          | LUNG-MAP; NCI-MATCH; STAMPEDE                                          |
| Adaptive Design       | Pre-planned interim analyses allow modification of randomisation, sample size, or arm dropping | I-SPY2 (breast); BATTLE (lung)                                         |
| Platform Trial        | Perpetually enrolling; arms added/dropped; master protocol                                     | RECOVERY (COVID precedent); multiple oncology platforms emerging       |
| Seamless Phase II/III | Phase II and III combined; interim determines if expansion to Phase III proceeds               | Efficient for rare tumours with biomarker enrichment                   |

## 5. Common Pitfalls in Interpreting Oncology Trial Data

- Surrogate endpoint assumption: PFS improvement does not guarantee OS benefit — always seek mature OS data when available
- Non-proportional hazards: Immunotherapy trials frequently show early curve convergence (higher initial events) before separation — log-rank test may underestimate true benefit; inspect RMST and landmark analyses
- Post-progression therapy confounding: Extensive crossover or active salvage may dilute OS differences even when PFS differences are large
- Biomarker-unselected vs. selected: A drug may show no benefit in an unselected population but significant benefit in a biomarker-defined subgroup — subgroup analyses require pre-specification and replication
- Absolute vs. relative risk reduction: A 30% reduction in HR may translate to only 2-3 months absolute benefit in OS — always report absolute numbers (NNT) for clinical interpretation
- Industry sponsorship bias: Meta-analyses show that industry-sponsored trials are more likely to report positive results — seek independent replication of practice-changing trials

**CLINICAL PEARL:** *The highest level of evidence for clinical decision-making is a well-powered, pre-registered Phase III RCT with a pre-specified primary endpoint of OS or a validated surrogate, analysed ITT with appropriate handling of missing data and adjudicated endpoints.*

## 6. Regulatory Pathways — FDA Accelerated Approval

The FDA Accelerated Approval (AA) pathway (21 CFR 314 Subpart H) grants approval based on a reasonably likely surrogate (ORR, PFS) while confirmatory Phase III trials are conducted. The 2022 Omnibus Act empowered FDA to withdraw AA more efficiently if confirmatory trials fail. As of 2024, several oncology indications previously granted AA have been voluntarily withdrawn or formally rescinded (e.g., multiple pembrolizumab, atezolizumab indications) following negative or inconclusive confirmatory trials — reinforcing the importance of awaiting mature OS data.

■ **EVIDENCE GRADING:** Always assess the quality of evidence (GRADE framework; Oxford CEBM levels) and context (patient population, comparator, follow-up) before extrapolating trial results to individual patients. 'Statistically significant' does not automatically equate to 'clinically meaningful'.